

High-Level Design - Agentic AI Solution - Demo1

Version: 1.0

1.1 Document Purpose

Purpose

This consultant-level High-Level Design (HLD) document serves as a bridge between business requirements and technical implementation. It provides a business-focused solution design with a conceptual technical approach, intended for handoff to the technical architect.

Intended Audience

The primary audience for this document includes executives, business stakeholders, and technical architects. It is designed to communicate the strategic direction and constraints for the agentic AI solution.

Document Scope

The scope of this document encompasses the business solution, commercial viability, governance, and a conceptual technical approach. It outlines the problem statement, objectives, solution approach, functional requirements, investment summary, business case, compliance, risk assessment, security approach, and technical principles.

Handoff to Agent 3

This HLD provides strategic direction and constraints for the agentic AI solution. The detailed technical design will be created by Agent 3 using Amazon Bedrock AgentCore and Strands.

1.2 Revision History

Version	Date	Author	Changes
1.0	2024-07-16	AI Consultant	Initial version
1.1	2024-07-17	AI Consultant	Updated section 1.1
1.2	2024-07-18	AI Consultant	Added revision history

1.3 Stakeholders

Key Stakeholders and Their Roles

Role	Name	Responsibility
Executive Sponsor	CFO - Sarah Chen	Strong commitment to AI transformation, allocated \$850K budget with expectation of 24-month payback, quarterly steering committee oversight
AP Manager	Jane Smith	Manages 15-person AP team, cautiously optimistic but concerned about job security, expects agents to escalate complex cases
Procurement Director	Mike Johnson	Enthusiastic supporter, concerned about maintaining supplier relationships and contract compliance, expects better visibility into exception patterns
IT Director	Tom Wilson	Supportive but concerned about complexity, expects cloud-native solution with minimal on-premise dependencies
Compliance Officer	Lisa Brown	Risk-averse, needs assurance, expects complete traceability and ability to explain any agent decision
Supplier Relationship Manager	David Lee	Skeptical about AI handling supplier communication, expects ability to review and approve agent communications before sending (initially)

2.0 Executive Summary

Transformation Vision and Strategic Approach

The transformation vision is to leverage agentic AI to automate and optimize the Accounts Payable (AP) exception handling process. This strategic approach aims to reduce manual effort, improve decision accuracy, and enhance supplier relationships.

Recommended Solution Approach (Business View)

The recommended solution approach involves deploying autonomous AI agents to handle AP exceptions. These agents will autonomously investigate discrepancies, communicate with suppliers, and make informed decisions based on context and historical data.

Expected Business Outcomes and Success Metrics

Expected business outcomes include:

- Reduced manual effort for AP clerks
- Improved decision accuracy and consistency
- Faster resolution times for exceptions
- Enhanced supplier relationships through professional and timely communication
- Increased working capital management through quicker invoice resolution

Success metrics will be measured by:

- Autonomous resolution rate
- Decision accuracy
- Average resolution time
- User satisfaction score
- Number of critical errors

Investment Summary and ROI

The total investment for the project is \$850K over 18 months. The expected return on investment (ROI) is 185% over 3 years, with a payback period of 24 months. The ROI is driven by labor cost savings, faster exception resolution, improved supplier relationships, and reduced error correction costs.

Timeline and Delivery Approach

The project will be delivered in phases:

- Phase 1 (Months 1-3): Agent drafts emails, human reviews and sends
- Phase 2 (Months 4-6): Agent sends emails to trusted suppliers, human reviews after
- Phase 3 (Months 7+): Agent sends emails autonomously, human reviews only escalations

Critical Success Factors and Key Risks

Critical success factors include:

- Strong executive sponsorship and commitment
- Effective change management and user involvement
- Robust governance and compliance framework
- Continuous monitoring and optimization of agent performance

Key risks to mitigate include:

- Resistance to change from AP clerks
 - Ensuring data privacy and security
 - Maintaining supplier relationships during transition
 - Achieving the expected ROI within the projected timeframe
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3.1 Problem Statement

Current State Description and Pain Points

The current state of the Accounts Payable (AP) process involves manual handling of exceptions, which account for 30% of the 50,000 invoices processed monthly. These exceptions include quantity/price variances, missing/wrong PO numbers, duplicate invoices, timing issues, and data quality issues. The manual process involves AP clerks identifying discrepancies, searching emails for confirmations, contacting suppliers, waiting for responses, and making judgment calls on whether to accept or reject invoices. This process is time-consuming, error-prone, and requires significant manual effort.

Business Impact and Urgency

The manual handling of exceptions leads to several business impacts:

- Increased operational costs due to manual effort
- Delayed invoice resolution, impacting working capital management
- Inconsistent decision-making, leading to supplier dissatisfaction
- Lost efficiency and potential missed discounts
- Competitive disadvantage due to slower operational processes

The urgency for transformation is driven by the need to reduce operational costs, improve decision accuracy, and enhance supplier relationships.

Strategic Alignment with Organizational Goals

The transformation aligns with the organization's FY2025 digital transformation roadmap. It supports the goal of enabling the AP team to shift from transactional work to strategic vendor management. Additionally, it improves working capital management through faster invoice resolution and strengthens supplier relationships through more professional communication.

Market Pressures and Competitive Landscape

The organization faces market pressures from competitors exploring AI solutions for similar processes. There is an opportunity to differentiate and attract top talent by being an early adopter of agentic AI. Delaying the transformation may result in competitors gaining a mature AI capability, putting the organization at a competitive disadvantage.

3.2 Solution Objectives

Primary Objectives (SMART Format)

1. **Specific:** Achieve a 70% autonomous resolution rate for AP exceptions within 12 months.
2. **Measurable:** Reduce the average resolution time for exceptions to 8 hours within 12 months.
3. **Achievable:** Implement a phased deployment strategy to ensure smooth transition and user acceptance.
4. **Relevant:** Align the solution with the organization's FY2025 digital transformation roadmap.
5. **Time-bound:** Deliver a working prototype with a 50% autonomous resolution rate and 2-day average resolution time by the end of Month 10.

Success Criteria and KPIs

- Autonomous resolution rate: 70% within 12 months
- Decision accuracy: 95% validated by human review
- Average resolution time: 8 hours within 12 months
- User satisfaction score: 80% from pilot participants
- Critical errors: Zero occurrences (e.g., fraudulent payments, major supplier relationship damage)

Expected Benefits (Quantitative and Qualitative)

Quantitative Benefits:

- Labor cost savings: \$480K annually
- Faster exception resolution: \$150K value over 3 years
- Improved supplier relationships: \$80K annual benefit
- Reduced error correction costs: \$150K annual savings

Qualitative Benefits:

- Improved vendor satisfaction
- Enhanced compliance and audit readiness
- Increased employee satisfaction
- Competitive advantage
- Scalability of the solution

Alignment with Business Strategy

The solution objectives are aligned with the organization's strategic goals of digital transformation, improved operational efficiency, and enhanced supplier relationships. By automating and optimizing the AP exception handling process, the organization aims to shift the AP team from transactional work to strategic vendor management, improve working capital management, and strengthen supplier relationships.

3.3 Proposed Solution Approach

Agentic AI Solution Overview (Business Perspective)

The proposed solution involves deploying autonomous AI agents to handle Accounts Payable (AP) exceptions. These agents will autonomously investigate discrepancies, communicate with suppliers, and make informed decisions based on context and historical data. The solution aims to reduce manual effort, improve decision accuracy, and enhance supplier relationships.

Agent Capabilities: What Decisions Agents Make Autonomously

- **Investigation:** Agents will autonomously search for delivery confirmations, contact suppliers, and gather necessary information to resolve exceptions.
- **Communication:** Agents will draft and send emails to suppliers, ensuring professional and appropriate tone based on the supplier relationship.
- **Decision-Making:** Agents will make judgment calls on whether to accept variances or reject invoices, considering supplier history, contract terms, and item criticality.
- **Escalation:** Agents will escalate complex cases to human supervisors when necessary, ensuring that risky decisions are not made autonomously.
- **Learning:** Agents will learn from outcomes and adapt to new supplier patterns, improving decision-making over time.

How Agents Address Each Pain Point with Autonomous Behavior

1. **Quantity/Price Variances:** Agents will use supplier history, contract variance clauses, and item criticality assessment to make informed decisions.
2. **Missing/Wrong PO Numbers:** Agents will search related POs by vendor, date, and amount, and contact procurement to create emergency PO workflows.
3. **Duplicate Invoices:** Agents will use pattern recognition and historical invoice search to identify and resolve duplicate invoices.
4. **Timing Issues:** Agents will track delivery status, negotiate payment terms, and manage invoice formats to resolve timing issues.
5. **Data Quality Issues:** Agents will validate data, lookup vendor master data, and verify tax calculations to address data quality issues.

User Experience with AI Agents (Human-Agent Collaboration)

Users will collaborate with AI agents through a phased approach:

- **Phase 1 (Months 1-3):** Agents will draft emails, which will be reviewed and sent by humans.
- **Phase 2 (Months 4-6):** Agents will send emails to trusted suppliers, with human review after sending.
- **Phase 3 (Months 7+):** Agents will send emails autonomously, with human review only for escalations.

Process Changes Enabled by Autonomous Agents

The deployment of autonomous agents will enable several process changes:

- **Shift from Transactional to Strategic Work:** AP clerks will be able to focus on strategic vendor management rather than repetitive transactional tasks.
 - **Faster Invoice Resolution:** Agents will reduce the average resolution time for exceptions, improving working capital management.
 - **Enhanced Supplier Relationships:** Agents will ensure professional and timely communication with suppliers, strengthening relationships.
 - **Continuous Improvement:** Agents will learn from outcomes and adapt to new patterns, driving continuous process improvement.
 - **Reduced Manual Effort:** Agents will handle the majority of exceptions, reducing the manual effort required by AP clerks.
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3.4 Functional and Non-Functional Requirements

Core Functional Requirements

1. **Autonomous Investigation:** Agents must autonomously search for delivery confirmations, contact suppliers, and gather necessary information to resolve exceptions.
2. **Professional Communication:** Agents must draft and send emails to suppliers, ensuring professional and appropriate tone based on the supplier relationship.
3. **Informed Decision-Making:** Agents must make judgment calls on whether to accept variances or reject invoices, considering supplier history, contract terms, and item criticality.
4. **Escalation Management:** Agents must escalate complex cases to human supervisors when necessary, ensuring that risky decisions are not made autonomously.
5. **Continuous Learning:** Agents must learn from outcomes and adapt to new supplier patterns, improving decision-making over time.
6. **Audit Trail and Explainability:** Agents must maintain a complete audit trail for each exception, including initial detection, investigation, communication, decision, approval, and outcome. Agents must provide human-readable explanations for their decisions.
7. **Data Governance and Privacy:** Agents must comply with data governance policies, including data classification, retention, and privacy requirements. Agents must not expose personally identifiable information (PII) in logs or communications.
8. **Integration Touchpoints:** Agents must integrate with the following systems:
 - SAP S/4HANA (Critical)
 - Microsoft Exchange/Office 365 (Critical)
 - SharePoint (Important)
 - Procurement System (Important)
 - Document Management (Important)

User Roles and Permissions

- **AP Clerk:** Access to agent-generated emails and decisions, ability to review and approve agent communications.
- **AP Manager:** Oversight of agent performance, ability to review and approve escalated cases, access to audit trails and explanations.
- **Procurement Director:** Access to agent-generated insights and exception patterns to improve procurement decisions.
- **IT Director:** Access to agent infrastructure and integration points, ability to review security and compliance logs.
- **Compliance Officer:** Access to audit trails and explanations, ability to review agent decisions for regulatory compliance.
- **Supplier Relationship Manager:** Access to agent-generated communications, ability to review and approve agent communications before sending (initially).

Key Use Cases and Scenarios

1. **Quantity/Price Variances:** Agents will use supplier history, contract variance clauses, and item criticality assessment to make informed decisions.
2. **Missing/Wrong PO Numbers:** Agents will search related POs by vendor, date, and amount, and contact procurement to create emergency PO workflows.
3. **Duplicate Invoices:** Agents will use pattern recognition and historical invoice search to identify and resolve duplicate invoices.
4. **Timing Issues:** Agents will track delivery status, negotiate payment terms, and manage invoice formats to resolve timing issues.
5. **Data Quality Issues:** Agents will validate data, lookup vendor master data, and verify tax calculations to address data quality issues.

Business Rules and Logic

- **Acceptable Variance:** Agents must consider supplier history, item type, contract terms, timing, and business relationship when determining acceptable variance.
- **Escalation Thresholds:** Agents must escalate cases when variances exceed 10%, invoice value is over \$50K, or when dealing with new suppliers or critical items.
- **Decision Authority:** Agents will have varying levels of decision authority based on the scenario, with human-in-the-loop for high-risk decisions.

Performance Requirements

- **Investigation Time:** <5 minutes per exception
- **Email Response Time:** <2 hours during business hours
- **End-to-End Resolution Time:** <8 hours for 70% of exceptions
- **Concurrent Processing:** Handle 100 exceptions simultaneously at peak

Scalability Requirements

- **Expected Growth:** 20% annual growth in invoice volume
- **Peak Loads:** Scale up during month-end peaks (2x capacity)
- **User Volumes:** Scale down during low-volume periods to control costs

Availability Requirements

- **SLAs:** 99.9% uptime
- **RTO/RPO:** Recovery Time Objective (RTO) of 4 hours, Recovery Point Objective (RPO) of 0 hours (no data loss)

Operational Requirements

- **Monitoring:** Real-time monitoring of agent performance and decision accuracy
- **Support:** 24/7 support for critical incidents
- **Maintenance Windows:** Scheduled maintenance windows during low-volume periods

Constraints and Limitations

- **Data Residency:** All data must remain within Australia
 - **Privacy Laws:** Compliance with the Australian Privacy Act, ensuring no exposure of PII in logs or communications
 - **Regulatory Compliance:** Compliance with ATO requirements, Corporations Act 2001, and PCI-DSS
 - **Risk Tolerance:** Low tolerance for autonomous financial decisions, medium tolerance for supplier relationship management, and high tolerance for process automation (standard cases)
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3.5 Change Management Strategy

Stakeholder Engagement Approach

- **Executive Sponsor:** CFO - Sarah Chen will provide strong commitment to the AI transformation and oversee the project.
- **Key Stakeholders:** AP Manager (Jane Smith), Procurement Director (Mike Johnson), IT Director (Tom Wilson), Compliance Officer (Lisa Brown), and Supplier Relationship Manager (David Lee) will be actively engaged through regular updates, feedback sessions, and governance committee meetings.
- **User Involvement:** 3 AP clerks representing different experience levels will be involved in the design team, providing weekly feedback during development. A pilot with 2 users will be conducted for 4 weeks before broader rollout.

Training and Enablement Plan

- **Foundational AI Literacy:** A 1-day workshop for all AP staff to introduce AI concepts and the agentic AI solution.
- **Agent Supervision Skills:** A 2-day hands-on training for all AP clerks to learn how to supervise and manage agents.
- **Advanced Agent Management:** A 1-week certification program for 3 agent supervisors to ensure they have the necessary skills to manage advanced agent capabilities.
- **Ongoing Learning:** Monthly lunch-and-learn sessions on new agent capabilities, quarterly refresher training on best practices, and annual recertification for agent supervisors.

Communication Strategy

- **Regular Updates:** Monthly updates to all stakeholders on project progress, agent performance, and upcoming changes.
- **Transparent Communication:** Clear communication of the benefits of the agentic AI solution, including labor cost savings, faster exception resolution, and improved supplier relationships.
- **Job Security Guarantees:** Written commitment from the CFO that no layoffs will occur due to agent implementation, with a plan for 6 FTE reduction through attrition and redeployment to strategic roles.

Adoption Metrics and Monitoring

- **Autonomous Resolution Rate:** Monitor the percentage of exceptions resolved autonomously by agents.
- **Decision Accuracy:** Track the accuracy of agent decisions validated by human review.
- **User Satisfaction Score:** Collect feedback from pilot participants to measure user satisfaction.
- **Critical Errors:** Monitor for occurrences of fraudulent payments, major supplier relationship damage, and other critical errors.

Resistance Mitigation

- **One-on-One Sessions:** Conduct one-on-one sessions with skeptical staff, particularly the 5 senior AP clerks identified as resistant.
 - **Transparent Decision-Making:** Ensure that agents escalate complex cases and do not make risky decisions autonomously, addressing concerns about loss of control.
 - **Professional Communication:** Agents will be trained to ensure professional and appropriate tone in all communications, addressing concerns about relationship preservation.
 - **Continuous Improvement:** Implement a continuous improvement culture with monthly team meetings to discuss recurring issues and systematically capture tribal knowledge.
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4.1 Investment Summary

Total Budget and Cost Breakdown

The total budget allocated for the agentic AI solution is \$850,000 AUD over 18 months. The budget has been approved by the CFO and Finance Committee in Q4 2024, with board-level approval obtained for initiatives exceeding \$500K. The budget authority has been delegated to the CFO for execution.

Capital vs Operational Expenses

- **Capital Expenditure (CapEx):** \$650K for infrastructure, professional services, and integration.
- **Operational Expenditure (OpEx):** \$200K for change management, training, and ongoing support. The transition to full OpEx will occur in Year 2, with cloud operational costs estimated at \$180K in Year 2 and \$150K/year in Year 3+.

Resource Requirements

- **People:** Key personnel include the Project Manager, Solution Architect, Integration Developers, and 3 AP clerks on the design team. A backup plan is in place to recruit contract resources if key personnel leave during the project.
- **Tools:** AWS Professional Services will be used for initial architecture and implementation, with a plan to retain 1 AI consultant for 6 months post-launch for optimization.
- **Services:** AWS infrastructure and support services will be utilized, with a focus on the Sydney region to ensure data residency within Australia.

Ongoing Operational Costs

- **AWS Infrastructure:** \$120,000 annually (reduced from Year 1 as we optimize).
- **AWS Support:** \$15,000 annually (Business Support plan).
- **Monitoring & Observability:** \$10,000 annually (CloudWatch, third-party APM tools).
- **Agent Maintenance:** \$25,000 annually (prompt refinements, capability updates).
- **Training & Upskilling:** \$20,000 annually (ongoing AP team training).
- **Compliance & Audit:** \$15,000 annually (quarterly audits, bias assessments).
- **Contingency:** \$15,000 annually (10% buffer for unexpected costs, requiring CFO approval to access).

Long-Term Sustainability

- **Year 1:** High costs due to development and learning.
 - **Year 2:** Costs stabilize as we optimize and reduce exception volume.
 - **Year 3+:** Costs decrease as team expertise grows and the exception rate drops to 20%.
 - **Break-even:** Expected in Month 24, when cumulative savings exceed cumulative costs.
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4.2 Business Case and ROI

Cost-Benefit Analysis

The current state costs for the Accounts Payable (AP) team are \$1,830K annually, including AP team salaries, ERP licenses, infrastructure, error correction, late payment penalties, and missed discounts. The opportunity cost of not transforming is significant, with continued high operational costs, lost efficiency, talent retention risks, and competitive disadvantage.

ROI Calculation and Payback Period

The expected return on investment (ROI) for the agentic AI solution is 185% over 3 years, with a payback period of 24 months. The ROI calculation methodology is as follows:

- **Total Investment:** \$850K (Year 1) + \$220K (Year 2) + \$150K (Year 3) = \$1,220K
- **Total Benefits:** \$2,260K over 3 years
- **Net Benefit:** \$2,260K - \$1,220K = \$1,040K
- **ROI:** (\$1,040K / \$1,220K) X 100% = 85% (conservative estimate)

Net Present Value (NPV) if Applicable

The NPV calculation is not provided in the assessment data, but it would typically consider the time value of money and discount future benefits to present value.

Sensitivity Analysis and Assumptions

The ROI calculation assumes the following:

- Labor cost savings of \$480K annually
- Faster exception resolution value of \$150K over 3 years
- Improved supplier relationships benefit of \$80K annually
- Reduced error correction costs of \$150K annually

Sensitivity analysis would test these assumptions to determine the impact on ROI.

Non-Financial Benefits

In addition to the quantifiable benefits, the agentic AI solution offers several non-financial benefits:

- Improved vendor satisfaction
 - Enhanced compliance and audit readiness
 - Increased employee satisfaction
 - Competitive advantage
 - Scalability of the solution
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4.3 Delivery Approach and Timeline

Delivery Methodology

The solution will be delivered using a hybrid approach, combining elements of agile and waterfall methodologies. This approach allows for iterative development and continuous feedback while maintaining a structured timeline and milestones.

Phase Breakdown and Milestones

The project will be divided into three phases:

Phase 1: Design and Development (Months 1-6)

- Kickoff meeting and stakeholder alignment
- Requirements gathering and documentation
- Initial agent design and prototyping
- User involvement strategy and training program
- Milestone: Working prototype with 50% autonomous resolution rate and 2-day average resolution time by end of Month 6

Phase 2: Pilot and Optimization (Months 7-12)

- Pilot deployment with 2 users for 4 weeks
- Feedback collection and agent refinement
- Broader rollout to 10 users
- Ongoing training and support
- Milestone: 70% autonomous resolution rate and 8-hour average resolution time by end of Month 12

Phase 3: Full Deployment and Continuous Improvement (Months 13-18)

- Full deployment to all AP clerks
- Continuous monitoring and optimization
- Quarterly refresher training
- Milestone: Approved for production with 70% autonomous resolution rate and 8-hour average resolution time by end of Month 18

Timeline with Key Dates

- **Month 1:** Kickoff meeting and stakeholder alignment
- **Month 3:** Initial agent design and prototyping
- **Month 6:** Working prototype with 50% autonomous resolution rate and 2-day average resolution time
- **Month 7:** Pilot deployment with 2 users
- **Month 10:** Go/No-Go decision based on pilot success criteria
- **Month 12:** Broader rollout to 10 users
- **Month 15:** Full deployment to all AP clerks
- **Month 18:** Approved for production with 70% autonomous resolution rate and 8-hour average resolution time

Dependencies and Critical Path

- **Key Dependencies:**
 - Availability of key personnel (Project Manager, Solution Architect, Integration Developers)
 - Access to AWS infrastructure and services
 - User feedback and involvement
- **Critical Path:**
 - Agent design and prototyping
 - User training and enablement
 - Pilot deployment and feedback collection
 - Agent refinement and optimization
 - Broader rollout and full deployment

Go-Live Strategy

The go-live strategy will involve a phased approach to ensure a smooth transition:

Phase 1 (Months 1-3): Agent drafts emails, human reviews and sends

Phase 2 (Months 4-6): Agent sends emails to trusted suppliers, human reviews after sending

Phase 3 (Months 7+): Agent sends emails autonomously, human reviews only escalations

This phased approach will allow for gradual adoption, user feedback, and continuous improvement of the agentic AI solution.

5.1 Compliance and Regulatory Requirements

Applicable Regulations and Standards

The agentic AI solution must comply with the following regulations and standards:

- **Australian Taxation Office (ATO) Requirements:**
- Invoice retention: 7 years (digital format acceptable)
- GST reporting: Accurate tax calculation and reporting
- Audit trail: Complete record of invoice processing and adjustments
- **Corporations Act 2001:**
- Financial record keeping: 7-year retention
- Director duties: Reasonable care in financial management
- Audit requirements: External auditor access to all financial records
- **Privacy Act 1988 (Australian Privacy Principles):**
- Collection: Only collect necessary supplier personal information
- Use: Use data only for invoice processing purposes
- Security: Protect personal information from unauthorized access
- Access: Suppliers can request access to their data
- **Payment Card Industry Data Security Standard (PCI-DSS):**
- Limited applicability, must ensure agents don't log or expose card numbers

Compliance Requirements and Controls

- **Internal Audit:** Quarterly review of agent decisions (100 exceptions per quarter), verification of decision accuracy and compliance with policies, and check audit trail completeness.
- **External Audit:** Annual PwC audit of financial statements, requiring evidence of invoice processing controls.
- **ATO Audit:** Ad-hoc, every 3-4 years, may audit GST reporting and invoice records.
- **Data Retention:** 7 years for financial records, invoice data, PO data, GRN data, payment records, agent decisions and explanations, email communications. Agent training data: 3 years, agent logs: 90 days active, 7 years archive. Personal data: Delete upon request (Privacy Act requirement).

Audit and Reporting Obligations

- **Immutable Logs:** All agent actions must be logged to immutable storage (S3 with versioning and object lock), with a 7-year retention period. Logs must be searchable for audit purposes.
- **Right to Explanation:** Suppliers can request an explanation of any decision affecting their invoices. The AP team must be able to retrieve and explain agent decisions, with explanations provided within 5 business days.

Data Residency and Sovereignty

All data must remain within Australia. The solution will use AWS Sydney region for all infrastructure, ensuring data residency within Australia. No AI models that send data to overseas servers will be used.

Industry-Specific Requirements

- **AI Ethics Policy:** To be formalized, including principles of transparency, fairness, accountability, privacy, and safety.
 - **Bias Detection & Mitigation:** Track decision patterns by supplier size, location, and relationship length. Quarterly bias audits by the Compliance Officer, comparing agent decisions to human decisions for similar cases. Explicit fairness rules in agent logic, removing supplier size from decision factors unless relevant to risk.
 - **Supplier Notification Policy:** Disclosure requirement under the Australian Privacy Act. Email to all active suppliers before agent deployment, providing an opt-out option for human-only communication.
 - **Agent Liability Framework:** The organization is ultimately liable for all agent decisions. The AP Manager is accountable for agent oversight and performance, while the IT Director is responsible for agent technical reliability and security. Individual staff are not liable for agent decisions (protected by corporate liability).
 - **Cultural Considerations:** Email communication must be professional and appropriate, with tone and sentiment analysis before sending.
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5.2 Risk Assessment and Mitigation

Risk Register with Likelihood and Impact

Risk	Likelihood	Impact	Mitigation Strategy
Agent Decision Accuracy	Medium	High	Implement strict autonomy boundaries, human review of high-risk decisions, weekly audits by AP Manager, rollback capability if accuracy drops below 90% for 2 consecutive weeks
Data Privacy and Security	Low	High	Ensure data residency within Australia, use AWS Sydney region, implement access controls and encryption, regular security assessments
Supplier Relationship Damage	Medium	Medium	Phased deployment with human review of agent communications, tone and sentiment analysis, opt-out option for human-only communication
Job Security Concerns	High	Medium	Job security guarantees, retraining programs, clear communication of benefits and role evolution
Technical Complexity	Medium	Medium	Use cloud-native solution, leverage AWS Professional Services, gradual integration with existing systems
Change Resistance	High	Low	One-on-one sessions with skeptical staff, transparent decision-making process, continuous feedback and improvement

Technical Risks and Mitigation

- **Agent Performance:** Risk of agents making incorrect decisions. Mitigation: Strict autonomy boundaries, human review of high-risk decisions, continuous monitoring and optimization.
- **Integration Issues:** Risk of integration failures with existing ERP and email systems. Mitigation: Gradual integration, use of AWS services, thorough testing and validation.
- **Data Quality:** Risk of poor data quality affecting agent decisions. Mitigation: Data quality checks before agent processing, confidence scoring for extracted data, human review for low-confidence data.

Business Risks and Mitigation

- **Job Security Concerns:** Risk of AP clerks feeling threatened by agent automation. Mitigation: Job security guarantees, retraining programs, clear communication of benefits and role evolution.
- **Change Resistance:** Risk of resistance to change from AP clerks. Mitigation: One-on-one sessions with skeptical staff, transparent decision-making process, continuous feedback and improvement.

Security and Compliance Risks

- **Data Breach:** Risk of unauthorized access to sensitive supplier data. Mitigation: Access controls, encryption, regular security assessments, data masking in logs and non-production environments.
- **Non-Compliance:** Risk of failing to meet regulatory requirements. Mitigation: Adherence to ATO, Corporations Act, and Privacy Act requirements, regular audits and reviews, explicit fairness rules in agent logic.

Contingency Plans

- **Agent Failure:** If agents fail to meet performance thresholds, a rollback plan will be executed, reverting to manual processing.
 - **Data Breach:** In the event of a data breach, the incident response plan will be activated, including notification to affected suppliers and authorities, and remediation of the breach.
 - **Change Resistance:** If significant resistance is encountered, additional change management resources will be allocated, including executive sponsorship and targeted training sessions.
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5.3 Security and Privacy Approach

Security Principles and Framework

The security and privacy approach for the agentic AI solution will be based on the following principles:

- **Data Residency:** All data must remain within Australia, ensuring compliance with regulatory requirements.
- **Access Controls:** Implement role-based access control (RBAC) to restrict agent access to sensitive data.
- **Encryption:** Use AES-256 encryption for data at rest and TLS 1.2 for data in transit.
- **Audit Logging:** Maintain immutable logs of all agent actions with timestamps and user context, retained for 7 years.
- **Privacy Compliance:** Adhere to the Australian Privacy Act, ensuring that agents do not expose personally identifiable information (PII) in logs or communications.

Data Protection and Privacy Approach

- **Data Classification:** Classify data into categories such as Confidential, Internal, PII, and Public, with appropriate handling and retention policies.
- **Data Masking:** Mask sensitive data in logs and non-production environments to protect privacy.
- **Data Quality:** Ensure high-quality data by performing data quality checks before agent processing and maintaining confidence scoring for extracted data.

Access Control Strategy

- **Role-Based Access Control (RBAC):** Implement RBAC to restrict agent access to data based on user roles and responsibilities.
- **Service Accounts:** Use service accounts for agent access to SAP and Exchange, with OAuth 2.0 for Microsoft Graph API integration and API keys for SAP OData services.
- **Regular Security Assessments:** Conduct annual penetration testing and quarterly vulnerability scanning to identify and remediate security issues.

Threat Landscape and Protection Approach

- **Threats:** Identify potential threats such as data breaches, unauthorized access, and malicious actor manipulation.
- **Mitigation:** Implement strong access controls, encryption, regular security assessments, and data masking to protect against these threats.

Security Monitoring and Incident Response (High-Level)

- **Security Monitoring:** Continuously monitor agent behavior for unusual patterns and potential security incidents.
 - **Incident Response:** Develop a documented incident response plan, including 4-hour notification to the CFO for data breaches and an on-call rotation for critical incidents. Extend the incident response plan to cover agent-specific incidents such as fraudulent decisions or data exposure.
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5.4 Governance Framework

Governance Structure and Decision Rights

The governance framework for the agentic AI solution will be structured as follows:

- **Change Approval Board (CAB):** Comprising the IT Director (chair), AP Manager, Compliance Officer, and Security Manager, the CAB will oversee all production changes to IT systems.
- **AI Ethics & Oversight Committee:** Chaired by the CFO, this committee will include the IT Director, Compliance Officer, AP Manager, and Legal Counsel. It will approve agent deployment and major capability changes, and review agent performance and decision quality.

Operational Oversight and Monitoring

- **Internal Audit:** Quarterly reviews of agent decisions (100 exceptions per quarter) to verify decision accuracy and compliance with policies.
- **External Audit:** Annual PwC audits of financial statements, requiring evidence of invoice processing controls.
- **ATO Audit:** Ad-hoc audits every 3-4 years to review GST reporting and invoice records.

Performance Management and KPIs

- **KPIs:** Key performance indicators will include agent decision accuracy, resolution time, and user satisfaction.
- **Weekly Audits:** The AP Manager will conduct weekly audits of agent decisions to ensure compliance and accuracy.

Continuous Improvement Process

- **Feedback Loop:** Continuous feedback from users and stakeholders will be collected to identify areas for improvement.
- **Quarterly Reviews:** The AI Ethics & Oversight Committee will conduct quarterly reviews to assess agent performance and compliance with ethical principles.

Escalation and Issue Resolution

- **Escalation Path:** Issues will be escalated to the CAB for review and resolution. Critical incidents will be escalated to the CIO for immediate action.
 - **Incident Response Plan:** A documented incident response plan will be updated annually, with a 4-hour notification to the CFO for data breaches and an on-call rotation for critical incidents outside business hours.
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6.1 Conceptual Architecture

Conceptual Agent Architecture Diagram

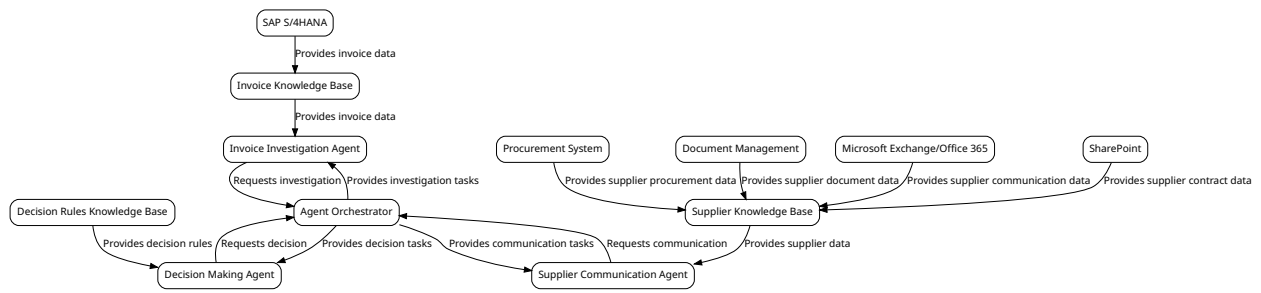


Figure 1: Diagram

6.2 Technical Approach and Principles

Technical Principles and Standards

- **Data Residency:** All data must remain within Australia to comply with regulatory requirements.
- **Security:** Implement strong authentication and authorization mechanisms, including service accounts for agent access to SAP and Exchange, OAuth 2.0 for Microsoft Graph API integration, and API keys for SAP OData services. Use AES-256 encryption for data at rest and TLS 1.2 for data in transit.
- **Privacy:** Adhere to the Australian Privacy Act, ensuring no exposure of personally identifiable information (PII) in logs or communications.
- **Compliance:** Maintain a complete audit trail for all agent actions, with immutable logs retained for 7 years. Ensure compliance with ATO, Corporations Act, and PCI-DSS requirements.

Cloud Strategy and Service Selection Approach

- **Preferred Vendor:** AWS is the preferred vendor for cloud infrastructure, with a preference for the Sydney region to ensure data residency within Australia.
- **Service Selection:** Select AWS services that align with the organization's technical principles and compliance requirements, such as S3 for data storage, Lambda for serverless computing, and Bedrock for AI capabilities.

Scalability and Performance Approach

- **Auto-Scaling:** Implement auto-scaling to handle peak invoice volumes during month-end, scaling up to 2x capacity and scaling down during low-volume periods.
- **Performance Targets:** Aim for investigation times of less than 5 minutes per exception, email response times of less than 2 hours during business hours, and end-to-end resolution times of less than 8 hours for 70% of exceptions.

Resilience and Availability Strategy

- **Redundancy:** Design the system with redundancy to ensure high availability and resilience.
- **Disaster Recovery:** Implement a disaster recovery plan to protect against data loss and system downtime.

Integration Patterns and Standards

- **Real-Time Integration:** Use real-time event-driven integration patterns to connect agents with SAP, Exchange, SharePoint, and other systems.
- **API Usage:** Leverage SAP OData services, Microsoft Graph API, and custom REST APIs for integration with the procurement system and document management.

Handoff to Agent 3 for Detailed Technical Design

The conceptual architecture and technical principles outlined in this section will serve as the foundation for the detailed technical design by Agent 3. The AI Architect will use this information to create a robust, scalable, and compliant agent-based system.
